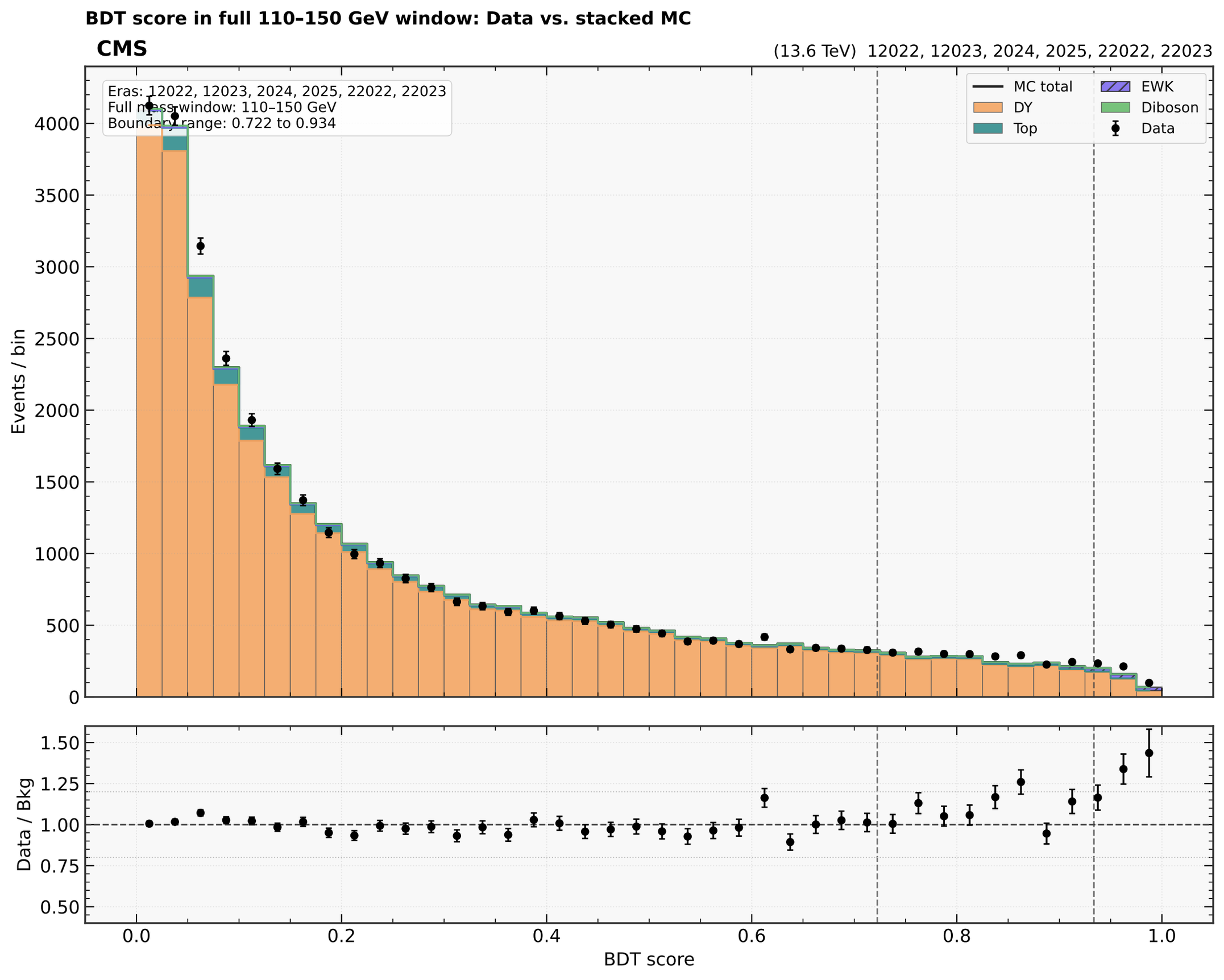
Run-3 H → μμ VBF BDT Category Optimisation

Eras : 12022, 12023, 2024, 2025, 22022, 22023
Category edges : 0.0000, 0.7225, 0.9337, 1.0000
MC scale factor : 1.0000 (Data/MC sideband normalisation)
Blinding window : 115–135 GeV
Sidebands : (110,115) ∪ (135,150) GeV
Outputs dir : /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Extra selection : –

CONTENTS:

p.2 BDT score distribution: Data vs. MC
p.3 BDT bin yields
p.4 Significance vs. number of categories
p.5 Boundary scan step-by-step (AN Fig. 32)
p.6 Atanh BDT + S/√B per category
p.7 BDT output vs. mass hypothesis (AN Fig. 26R)
p.8 Signal FWHM vs. BDT score
p.9 ROC curve with AUC
p.10 Blinded dimuon mass spectrum
p.11 Mass sculpting check (AN Fig. 25)
p.12 Per-category mass distributions and background fits
p.13 Background model comparison (χ²/ndf)
p.14 VBF signal DCB fits per category
p.15 DCB parameter table
p.16 σ_CB vs. category
p.17 Signal acceptance × efficiency
p.18 Per-category yield and sensitivity summary
p.19 Signal mass shapes mH=120/125/130 (AN Fig. 40)
p.20 Category summary table
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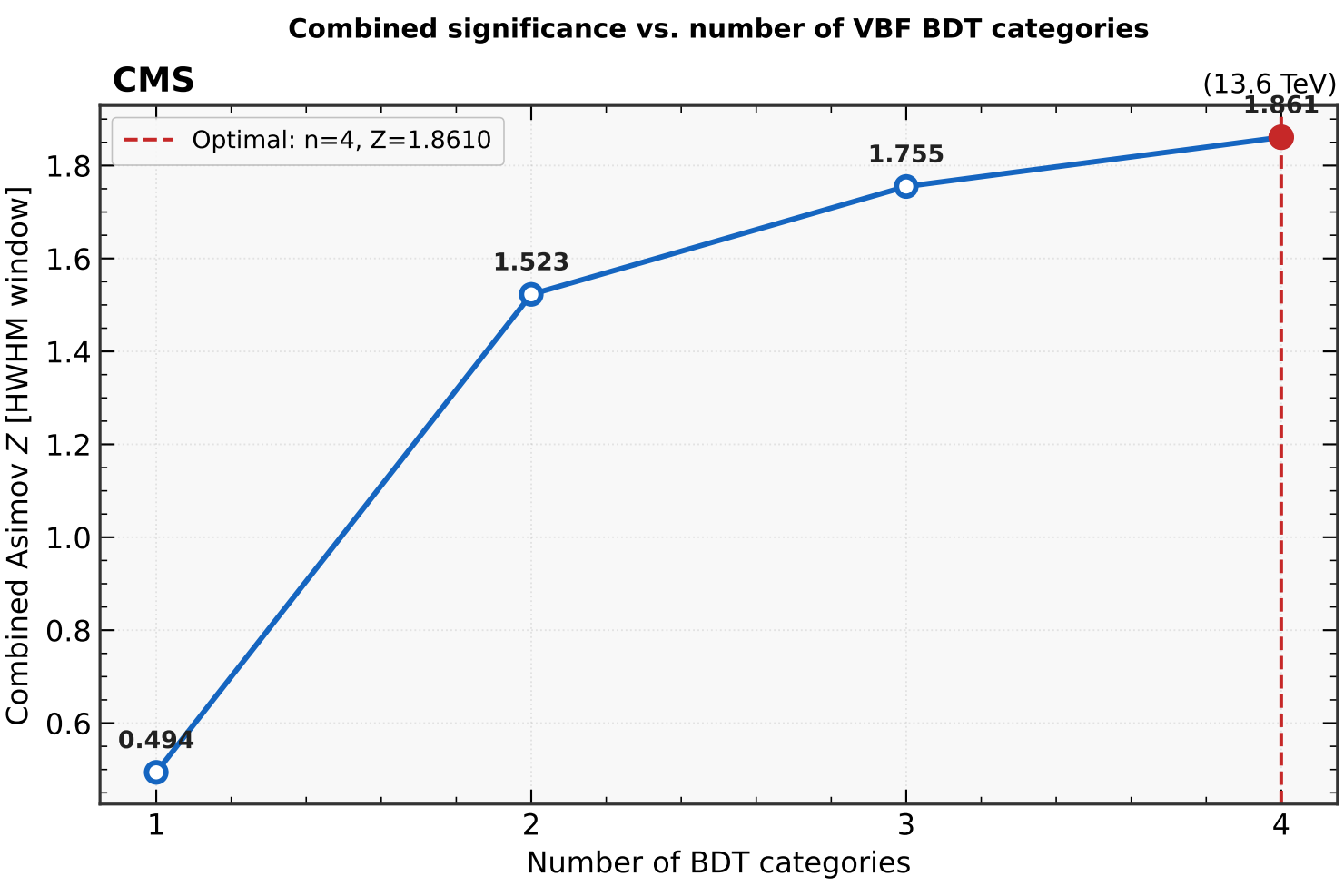

BDT score in full 110–150 GeV window: Data vs. stacked MC
CMS
(13.6 TeV) 12022, 12023, 2024, 2025, 22022, 22023
Eras: 12022, 12023, 2024, 2025, 22022, 22023
Full mass window: 110–150 GeV
Boundary range: 0.722 to 0.934
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data / Bkg
BDT score

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

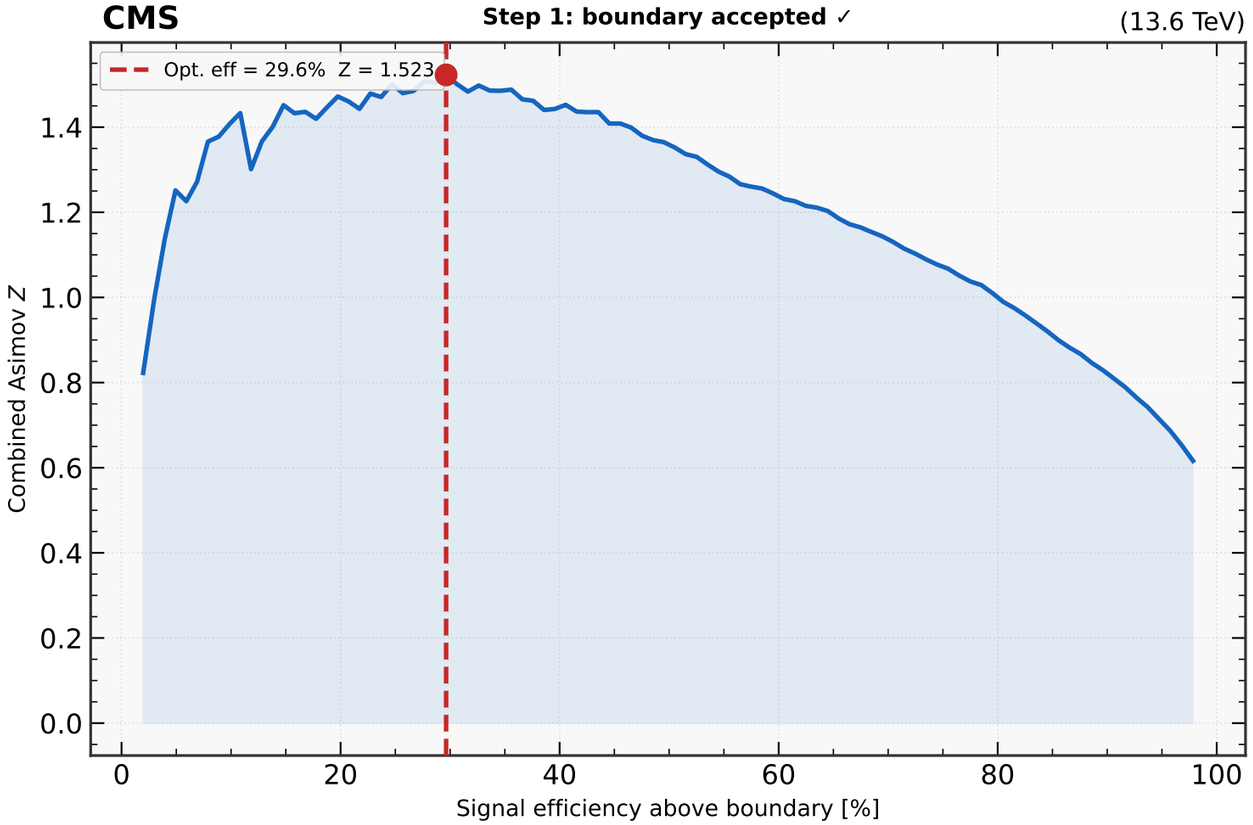
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.000, 0.025)	0.176	4101.23 ± 33.58	4124	0.00
[0.025, 0.050)	0.371	3984.33 ± 34.89	4051	0.01
[0.050, 0.075)	0.429	2935.25 ± 30.21	3145	0.01
[0.075, 0.100)	0.484	2298.68 ± 26.91	2361	0.02
[0.100, 0.125)	0.494	1888.97 ± 24.70	1931	0.03
[0.125, 0.150)	0.513	1616.91 ± 22.91	1591	0.03
[0.150, 0.175)	0.529	1349.33 ± 20.95	1372	0.04
[0.175, 0.200)	0.565	1205.83 ± 20.07	1146	0.05
[0.200, 0.225)	0.577	1066.81 ± 18.89	996	0.05
[0.225, 0.250)	0.589	939.75 ± 17.72	933	0.06
[0.250, 0.275)	0.603	846.91 ± 16.52	826	0.07
[0.275, 0.300)	0.622	773.17 ± 15.91	763	0.08
[0.300, 0.325)	0.651	712.67 ± 15.30	664	0.09
[0.325, 0.350)	0.669	643.60 ± 14.41	633	0.10
[0.350, 0.375)	0.698	632.33 ± 14.27	593	0.11
[0.375, 0.400)	0.735	585.06 ± 13.86	602	0.13
[0.400, 0.425)	0.756	559.57 ± 13.73	564	0.13
[0.425, 0.450)	0.806	553.65 ± 13.84	530	0.15
[0.450, 0.475)	0.834	520.23 ± 13.20	505	0.16
[0.475, 0.500)	0.869	479.75 ± 12.76	474	0.18

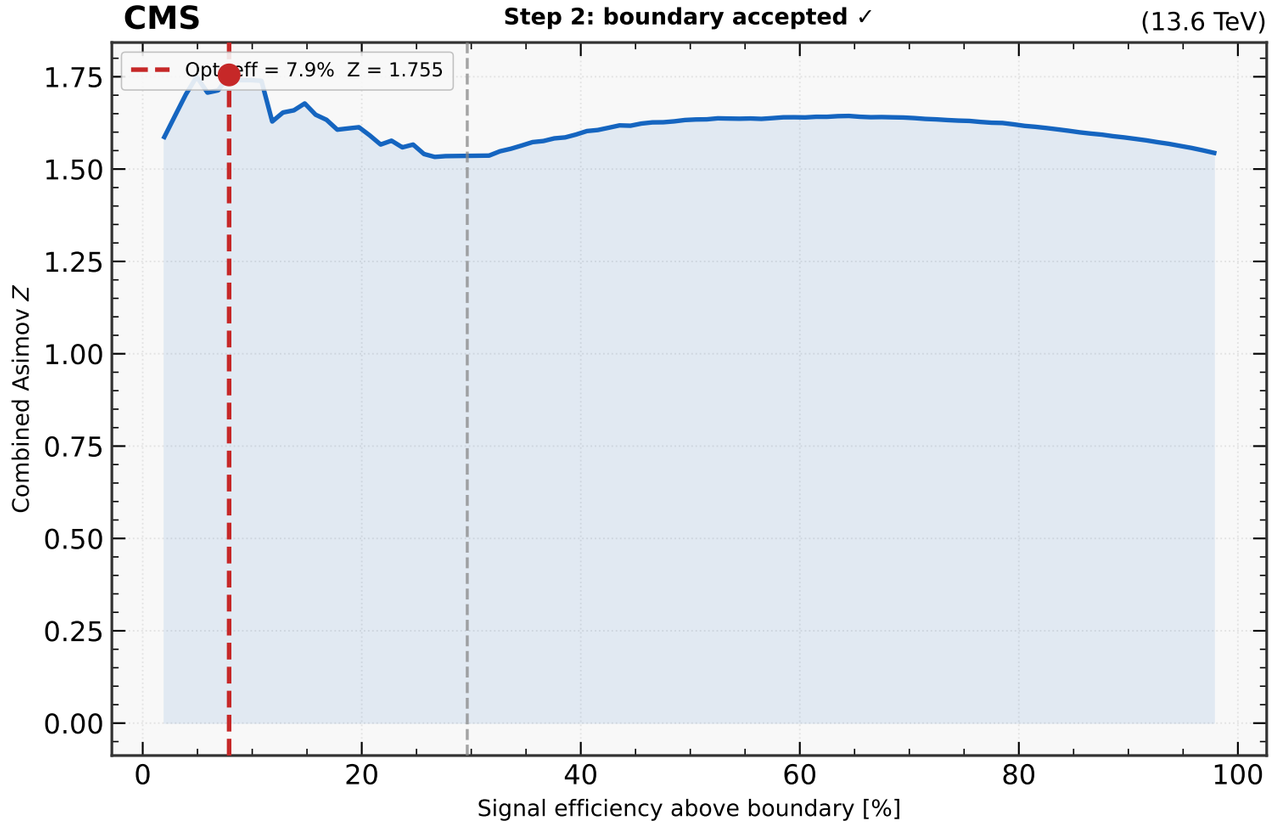
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.500, 0.525)	0.923	462.00 ± 12.38	443	0.20
[0.525, 0.550)	0.972	417.13 ± 11.62	387	0.23
[0.550, 0.575)	1.031	407.73 ± 11.28	393	0.25
[0.575, 0.600)	1.085	375.94 ± 10.94	369	0.29
[0.600, 0.625)	1.149	359.43 ± 10.58	418	0.32
[0.625, 0.650)	1.214	371.43 ± 11.36	332	0.33
[0.650, 0.675)	1.290	341.75 ± 10.33	342	0.38
[0.675, 0.700)	1.386	328.39 ± 10.06	337	0.42
[0.700, 0.725)	1.501	323.81 ± 10.35	328	0.46
[0.725, 0.750)	1.616	307.57 ± 10.03	309	0.52
[0.750, 0.775)	1.770	279.45 ± 9.24	316	0.63
[0.775, 0.800)	1.957	285.38 ± 9.52	300	0.68
[0.800, 0.825)	2.150	282.63 ± 9.66	299	0.75
[0.825, 0.850)	2.455	242.39 ± 8.79	283	1.00
[0.850, 0.875)	2.807	231.06 ± 8.30	291	1.20
[0.875, 0.900)	3.315	239.01 ± 8.75	226	1.37
[0.900, 0.925)	4.056	213.88 ± 8.21	244	1.86
[0.925, 0.950)	5.258	200.97 ± 7.97	234	2.55
[0.950, 0.975)	7.396	159.16 ± 6.76	213	4.44
[0.975, 1.000)	7.654	68.25 ± 4.07	98	10.08

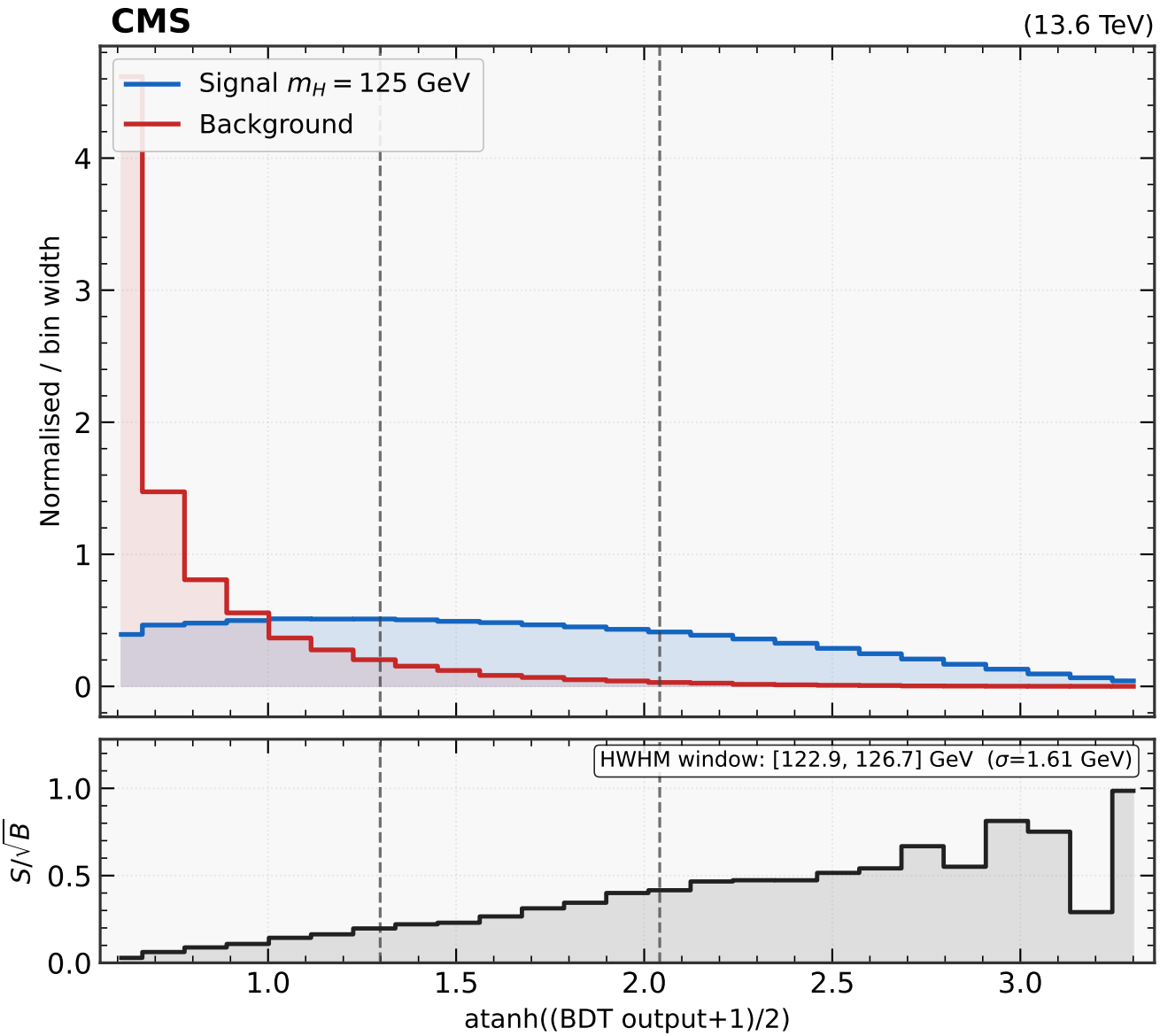

Combined significance vs. number of VBF BDT categories
CMS
(13.6 TeV)
Optimal: n=4, Z=1.8610
0.494
1.523
1.755
1.861
Combined Asimov Z [HWHM window]
Number of BDT categories
0.6
0.8
1.0
1.2
1.4
1.6
1.8
1
2
3
4

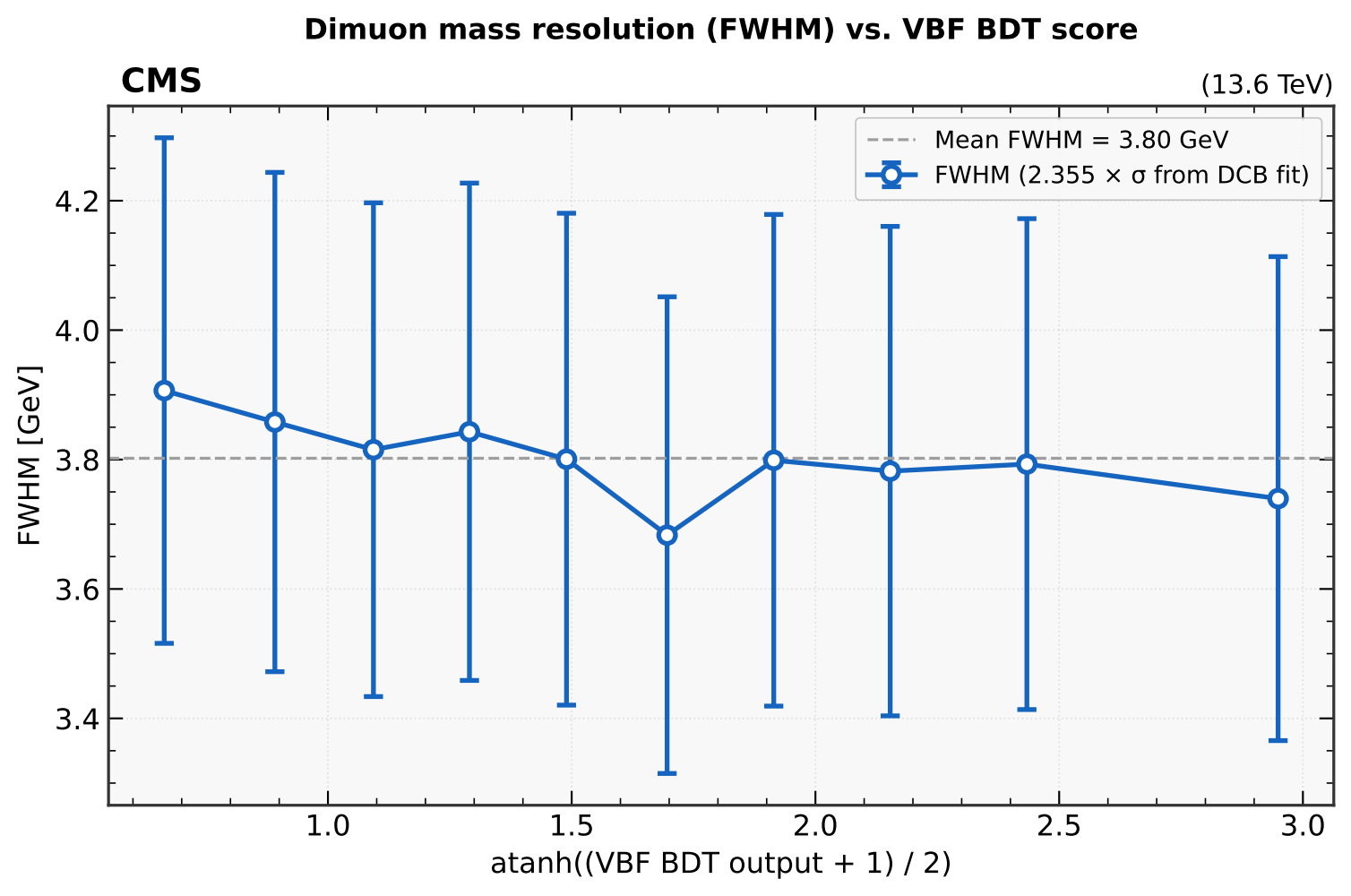
Iterative category boundary optimisation (AN Fig. 32 analog)

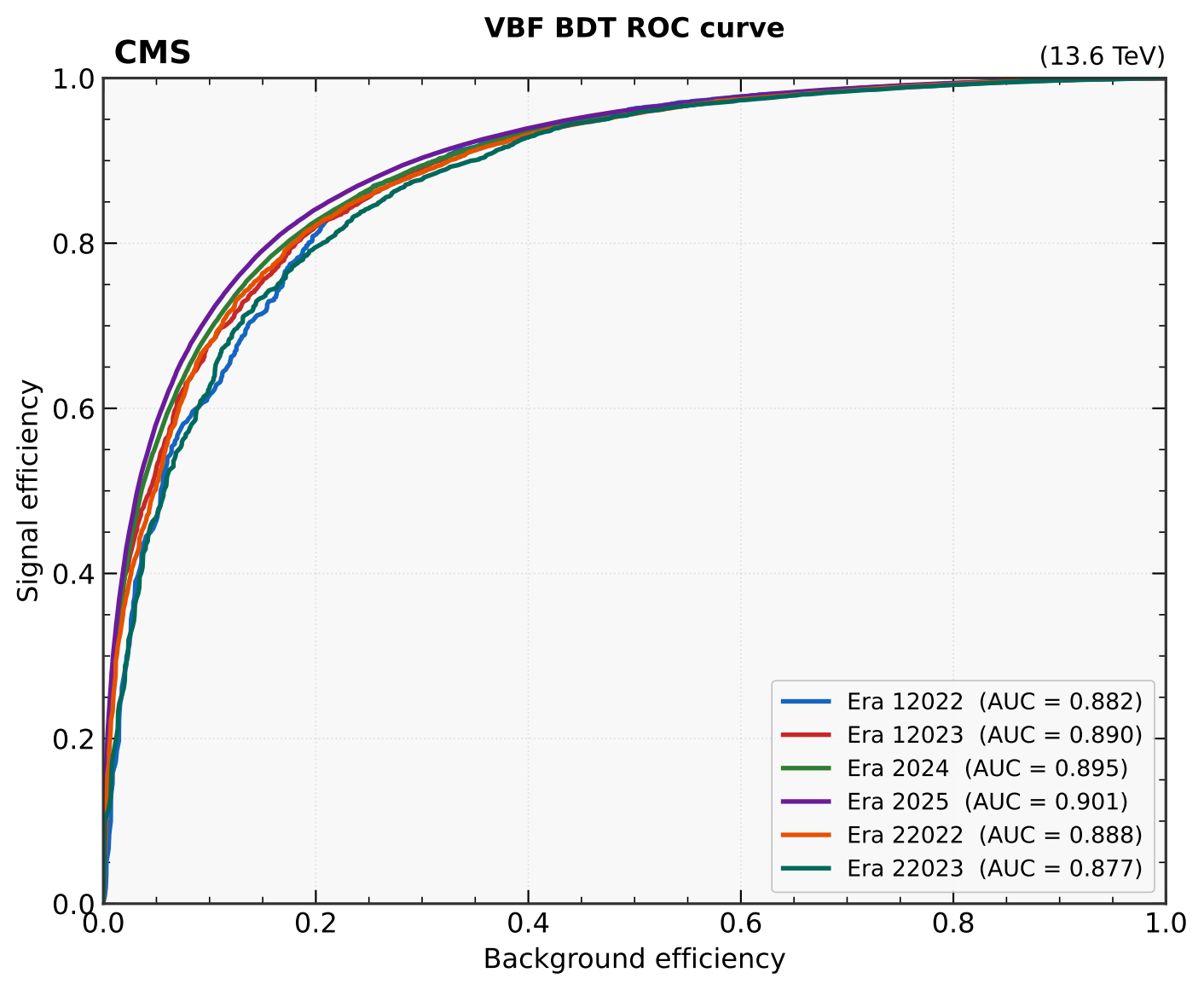


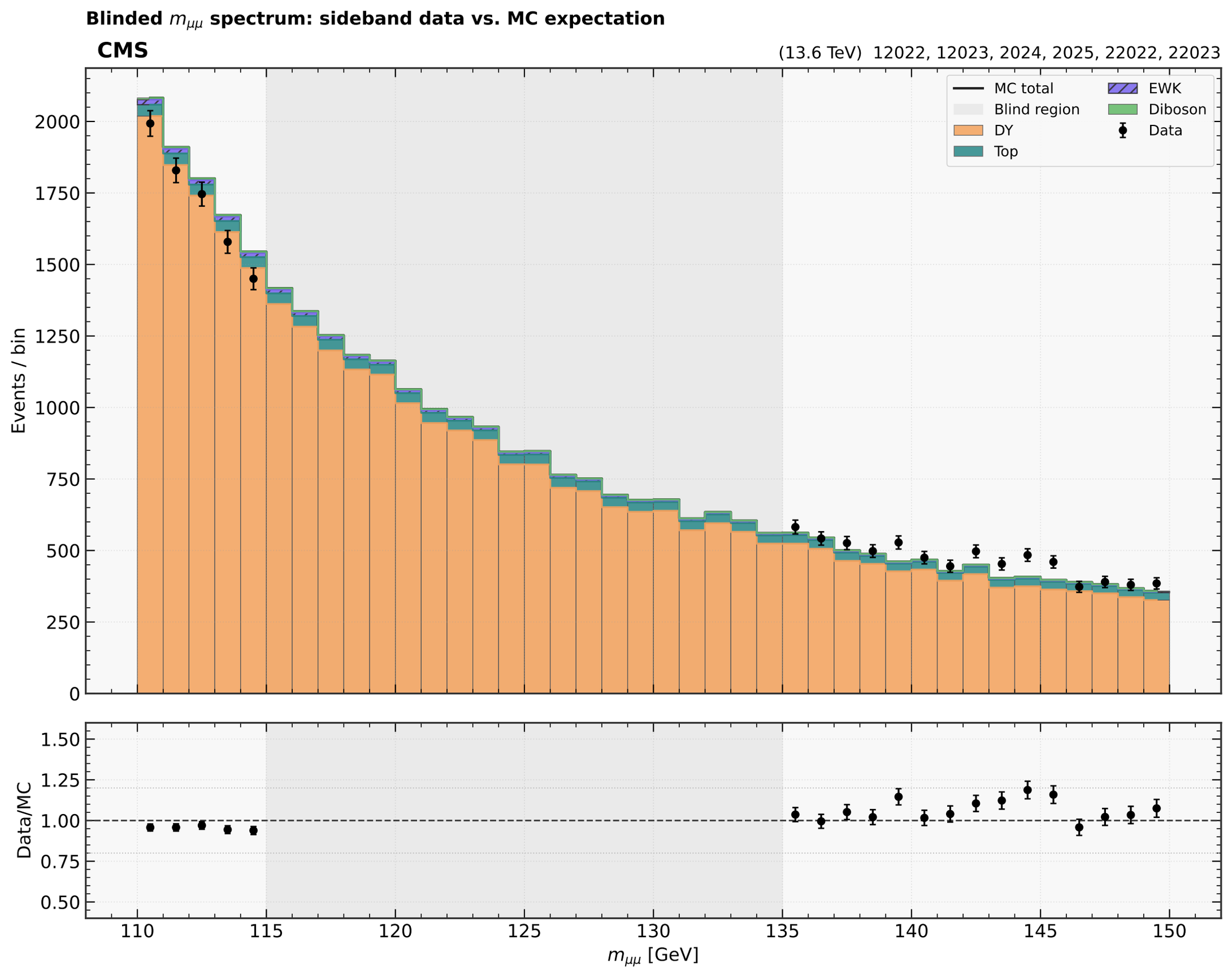


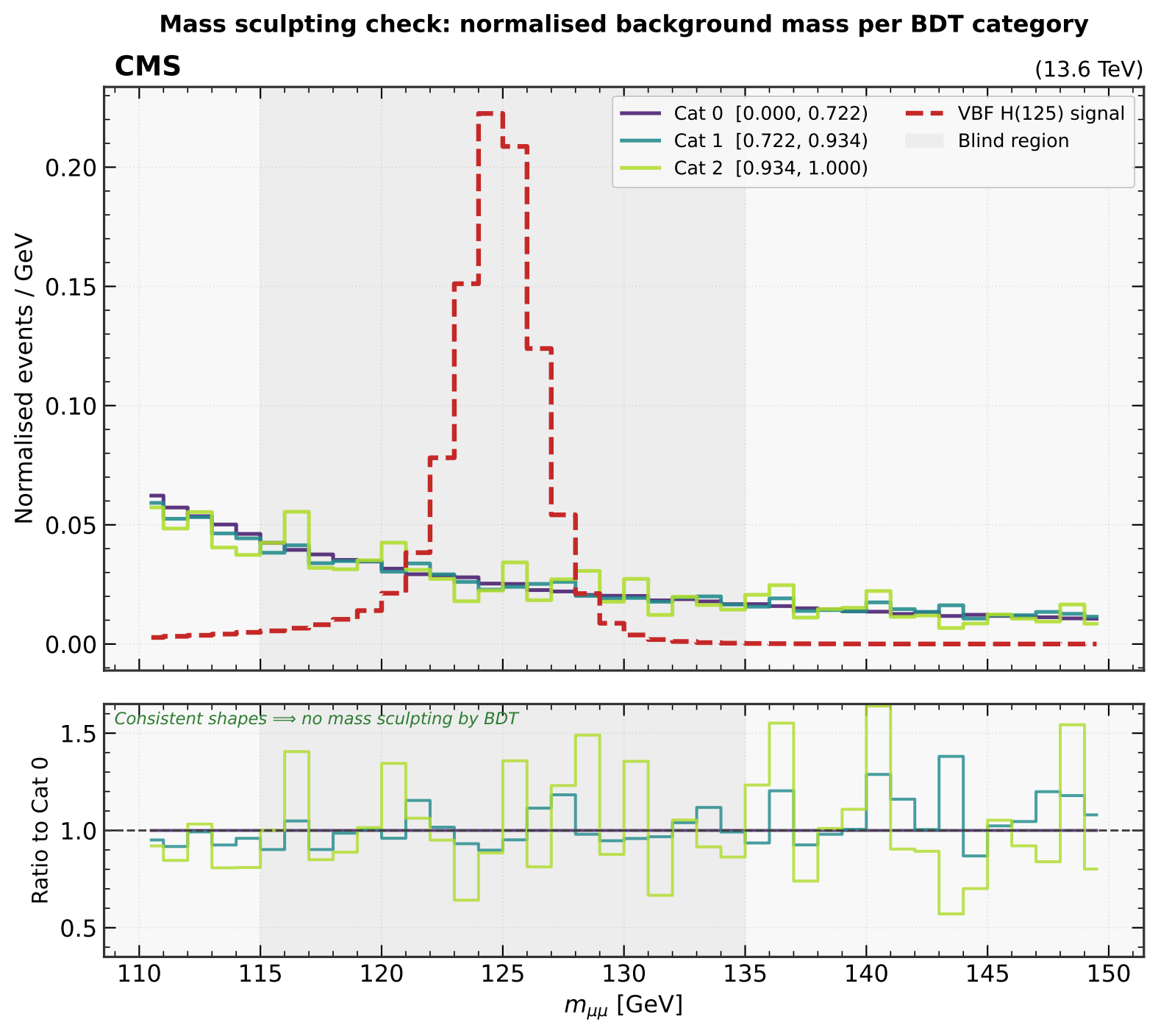
VBF BDT score and per-bin $S/\sqrt{B}$ in signal HWHM window



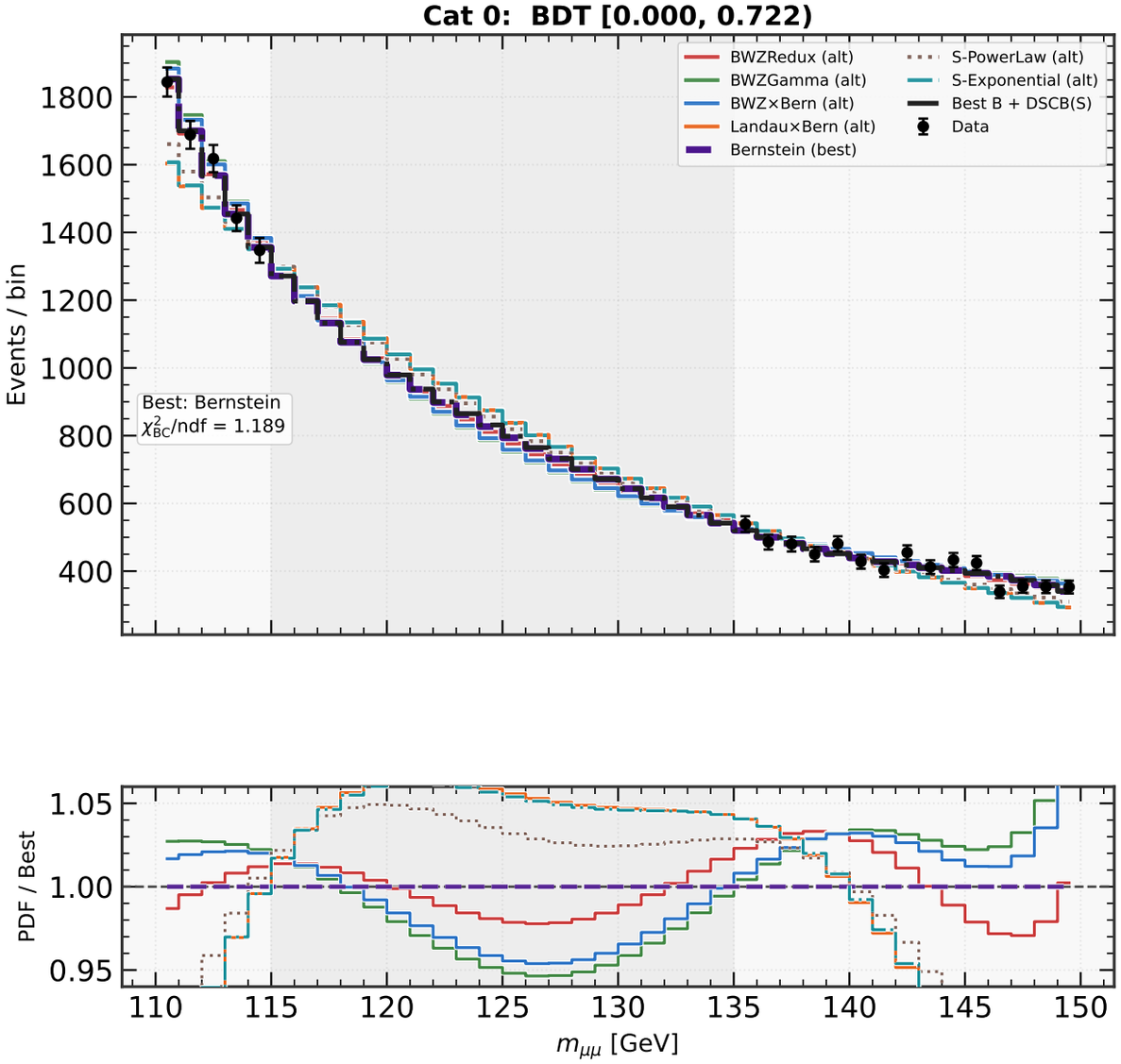

Dimuon mass resolution (FWHM) vs. VBF BDT score
CMS
(13.6 TeV)
Mean FWHM = 3.80 GeV
FWHM (2.355 × σ from DCB fit)
FWHM [GeV]
4.2
4.0
3.8
3.6
3.4
1.0
1.5
2.0
2.5
3.0
atanh((VBF BDT output + 1) / 2)

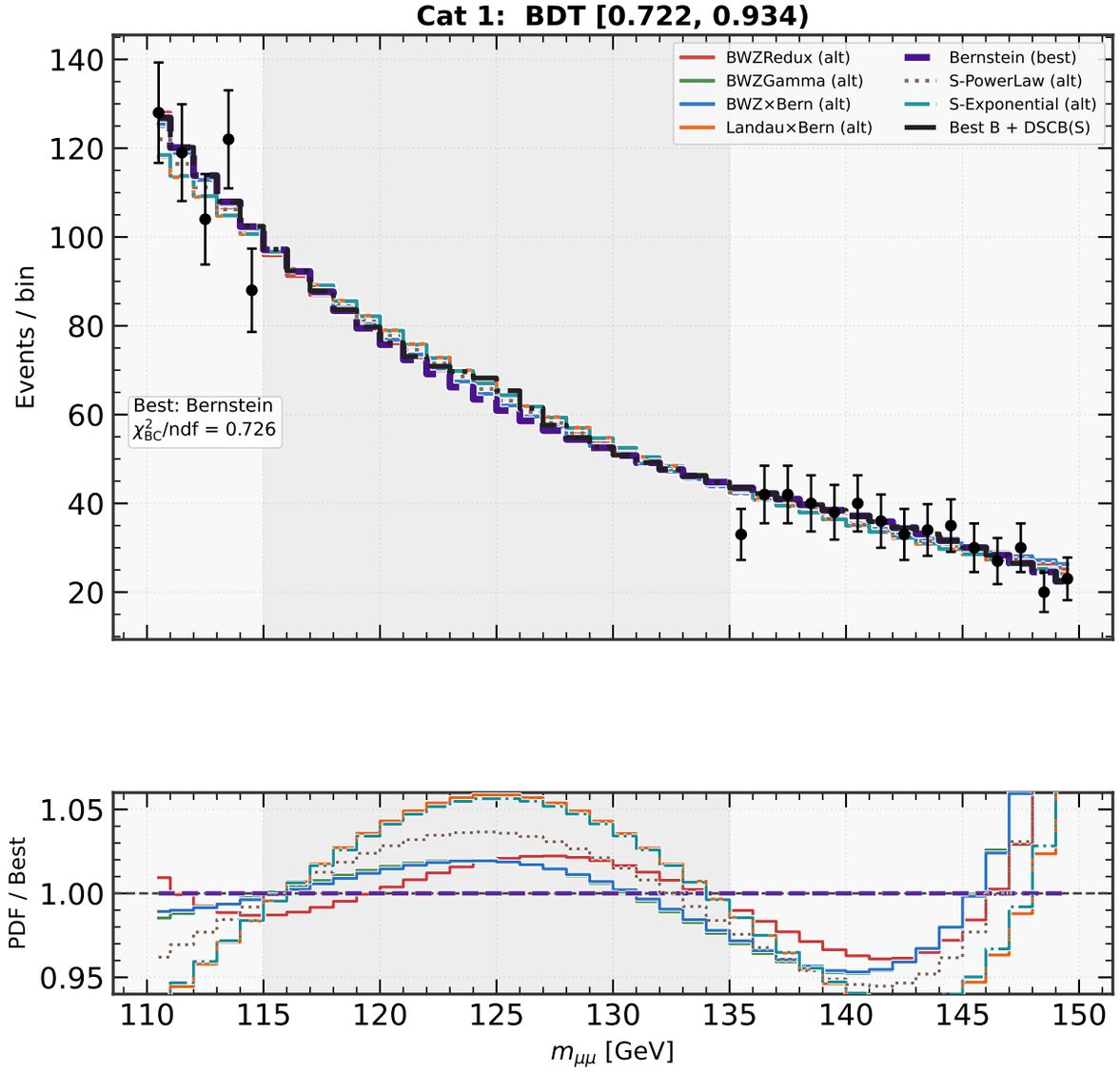

VBF BDT ROC curve
CMS
(13.6 TeV)
Signal efficiency
Background efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Era 12022 (AUC = 0.882)
Era 12023 (AUC = 0.890)
Era 2024 (AUC = 0.895)
Era 2025 (AUC = 0.901)
Era 22022 (AUC = 0.888)
Era 22023 (AUC = 0.877)

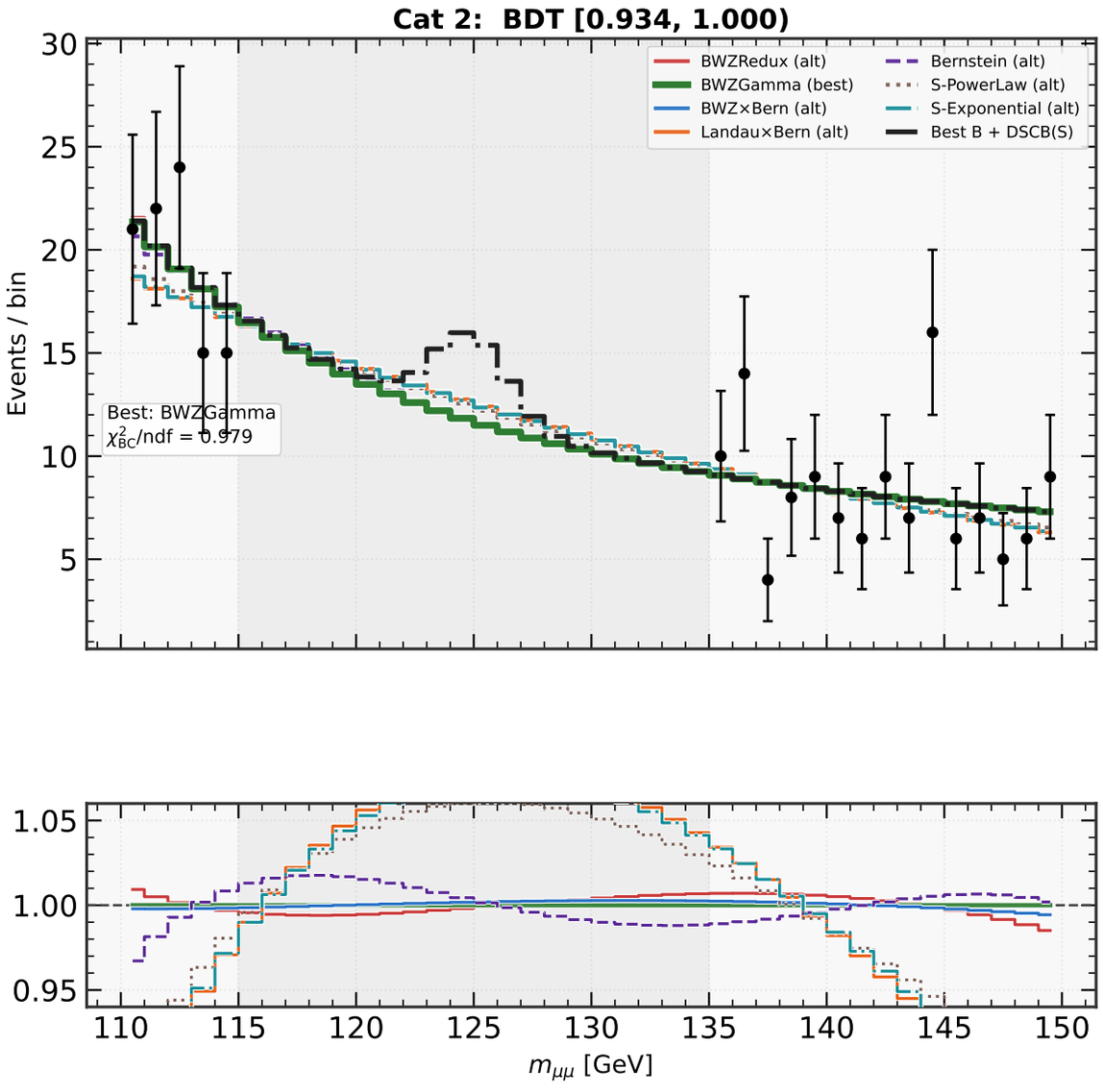

Blinded $m_{\mu\mu}$ spectrum: sideband data vs. MC expectation
CMS
(13.6 TeV) 12022, 12023, 2024, 2025, 22022, 22023
MC total
Blind region
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
$m_{\mu\mu}$ [GeV]


Mass sculpting check: normalised background mass per BDT category
CMS
(13.6 TeV)
Cat 0 [0.000, 0.722)
Cat 1 [0.722, 0.934)
Cat 2 [0.934, 1.000)
VBF H(125) signal
Blind region
Normalised events / GeV
0.00
0.05
0.10
0.15
0.20
Consistent shapes ⟹ no mass sculpting by BDT
Ratio to Cat 0
0.5
1.0
1.5
110
115
120
125
130
135
140
145
150
$m_{\mu\mu}$ [GeV]

Per-category dimuon mass: background model fits



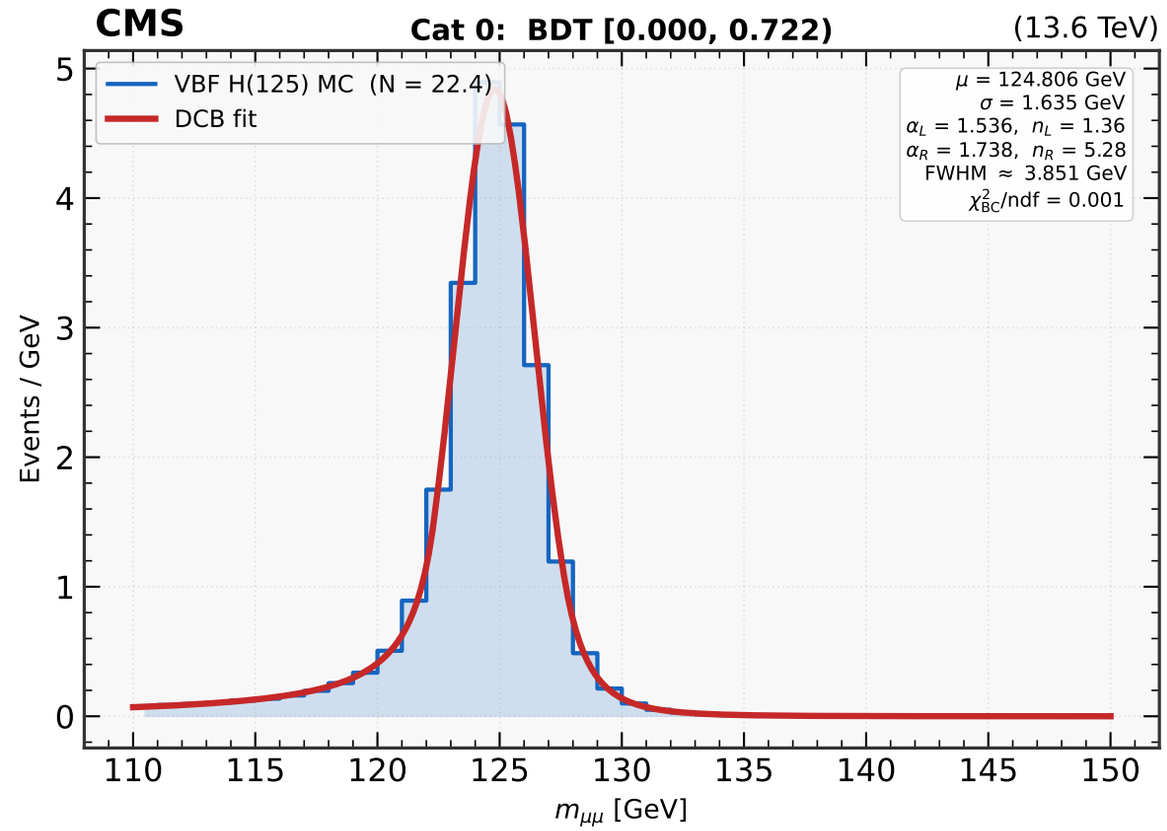


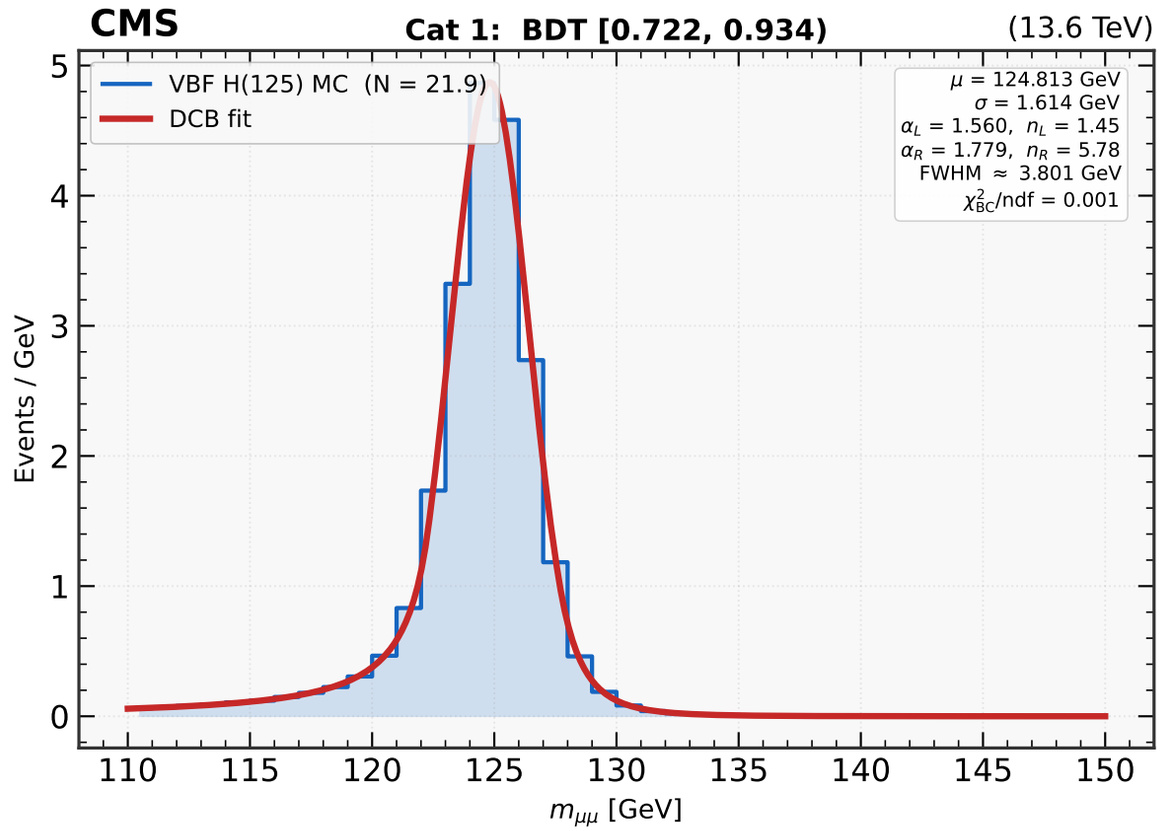


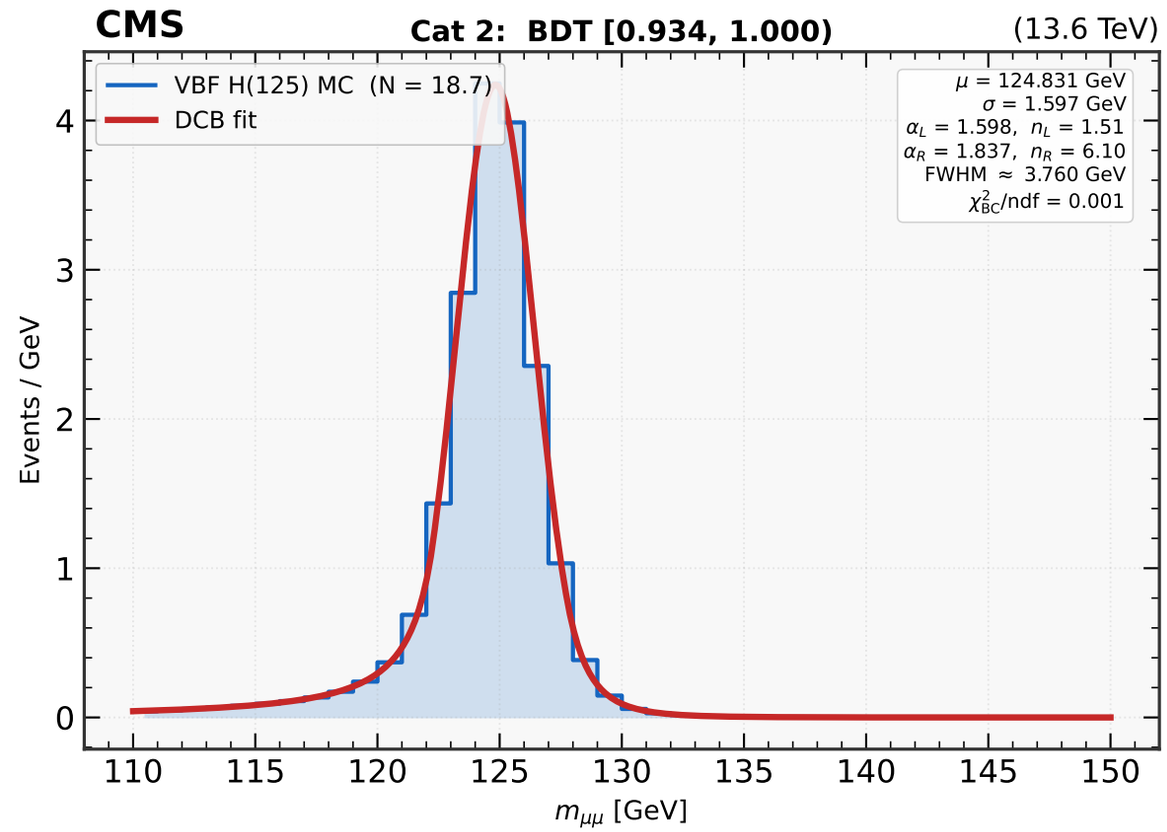
Per-category background model fit comparison (χ²_BC/ndf, ndf)

Category	BDT range	Best model	Best χ²_BC/ndf	BWZRedux	BWZGamma	BWZ×Bern	Landau×Bern	Bernstein	S-PowerLaw	S-Exponential
0	[0.000, 0.722)	Bernstein	1.189	1.349 (37)	2.007 (37)	1.778 (36)	5.265 (37)	1.189 (33)	3.288 (37)	5.080 (37)
1	[0.722, 0.934)	Bernstein	0.726	0.737 (37)	0.749 (37)	0.768 (36)	0.815 (37)	0.726 (35)	0.760 (37)	0.807 (37)
2	[0.934, 1.000)	BWZGamma	0.979	0.979 (37)	0.979 (37)	1.006 (36)	1.036 (37)	1.037 (35)	1.012 (37)	1.030 (37)

VBF H(125) signal DCB extraction per BDT category

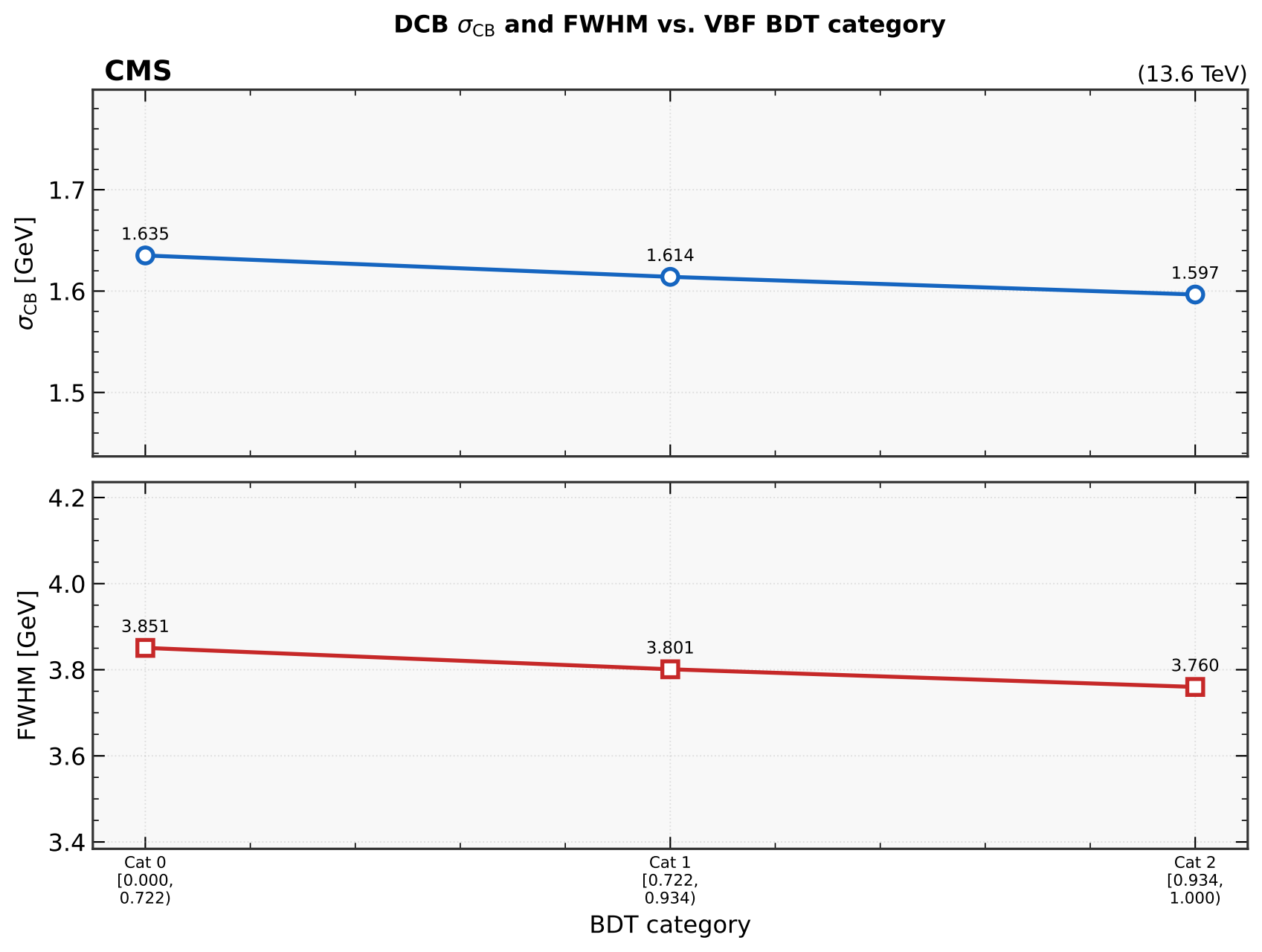


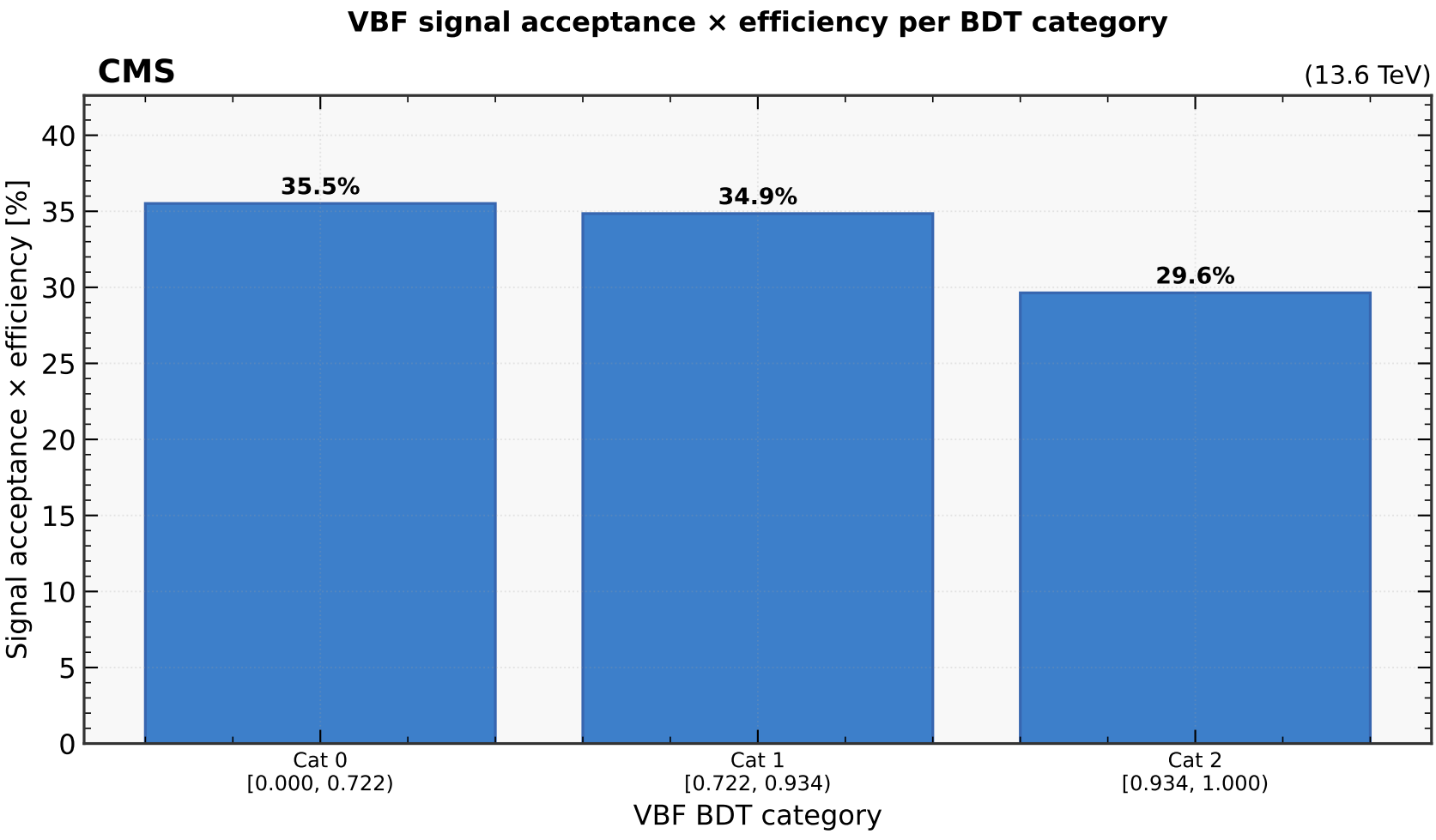




VBF signal DCB fit parameters per category

Category	BDT range	N_sig	Mean [GeV]	Sigma [GeV]	alpha_L	n_L	alpha_R	n_R	FWHM [GeV]	Asimov Z	χ^2_BC/ndf
0	[0.000, 0.722)	22.36	124.8061	1.6351	1.5365	1.357	1.7381	5.282	3.851	0.179	0.001
1	[0.722, 0.934)	21.94	124.8129	1.6140	1.5602	1.445	1.7790	5.775	3.801	0.686	0.001
2	[0.934, 1.000)	18.66	124.8306	1.5966	1.5976	1.508	1.8372	6.101	3.760	1.480	0.001


DCB σ_{CB} and FWHM vs. VBF BDT category
CMS
(13.6 TeV)
σ_{CB} [GeV]
1.7
1.6
1.5
1.635
1.614
1.597
FWHM [GeV]
4.2
4.0
3.8
3.6
3.4
3.851
3.801
3.760
Cat 0
[0.000,
0.722)
Cat 1
[0.722,
0.934)
Cat 2
[0.934,
1.000)
BDT category


VBF signal acceptance × efficiency per BDT category
CMS
(13.6 TeV)
Signal acceptance × efficiency [%]
0
5
10
15
20
25
30
35
40
35.5%
34.9%
29.6%
Cat 0
[0.000, 0.722)
Cat 1
[0.722, 0.934)
Cat 2
[0.934, 1.000)
VBF BDT category

Per-category yield and sensitivity summary

Category	BDT range	S_full	B_full	S_HWHM	B_HWHM	HWHM window [GeV]	S/√B	Asimov Z
0	[0.000, 0.722)	22.36	73258.90	15.266	7235.733	[122.9, 126.7]	0.179	0.179
1	[0.722, 0.934)	21.94	5161.59	15.118	481.046	[122.9, 126.7]	0.689	0.686
2	[0.934, 1.000)	18.66	832.41	13.026	73.278	[123.0, 126.7]	1.522	1.480

Category summary: HWHM-windowed signal and background yields

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	S/√B [HWHM]	Asimov Z (cat) [HWHM]	B_HWHM
0	[0.000, 0.722)	14331	34062.30 ± 158.73	0.421	73258.90 ± 233.44	0.1795	0.1794	7235.733
1	[0.722, 0.934)	1064	2409.01 ± 42.49	0.442	5161.59 ± 62.32	0.6893	0.6857	481.046
2	[0.934, 1.000)	220	369.49 ± 15.35	0.595	832.41 ± 23.48	1.5217	1.4797	73.278
Combined	all						1.6407	

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total_err2	total
0	[0.000, 0.722)	69774.07	354.81	2875.91	254.11	54496.41	73258.90
1	[0.722, 0.934)	4759.48	243.41	137.36	21.34	3883.58	5161.59
2	[0.934, 1.000)	648.05	160.47	21.14	2.75	551.35	832.41